Recommendations — aml-mds-related-rr-tp53-aberrant-hla-pending-x7q2

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

Generated 2026-07-10

Recommendations — aml-mds-related-rr-tp53-aberrant-hla-pendi

Profile snapshot

- **Primary site:** bone marrow (hematologic)
- **Histology:** Acute myeloid leukemia, myelodysplasia-related (AML-MR), relapsed/refractory; transformed from prior MDS (RCMD, IPSS-R high/very-high, diagnosed ~2019); with extramedullary myeloid sarcoma
- **Stage:** Relapsed/refractory AML-MR. Marrow ~85% blasts (June 2026); peripheral-blood blasts 58->82% (late June 2026); extramedullary right-thigh myeloid sarcoma on MRI (May 2026, adverse feature). Refractory to FLAG-IDA + venetoclax reinduction. Late relapse ~6 years after matched-sibling PBSC allo-HCT (~2020); pattern consistent with graft-versus-leukemia escape rather than graft failure.
- **ECOG:** 1
- **Age band:** 50-59
- **Sex:** female
- **Biomarkers:** TP53 aberrant, pending confirmation (NOT resolved in either direction). Pathogenic TP53 mutation reported on 2019 diagnostic NGS; 2026 relapse NGS reported NEGATIVE, but on a limited assay (LOD ~2% VAF, no copy-number/CNV detection, not MRD-grade). Discrepancy most plausibly subclonal loss, copy-loss, or a re-review artifact rather than true reversion. Working call: TP53-aberrant pending confirmation. (ngs_pending); TP53 Y220C (specific variant) unknown / to be determined — Y220C status not established; therapy handle (rezatapopt) is contingent on it (ngs_pending); HLA-A*02:01 unknown / pending retrieval. High-resolution HLA typing was performed ~2020 but the allele-level result is not in the records on hand (placeholder in the EHR). (unknown); HLA-A*24:02 unknown / pending (unknown); HA-1 / HA-2 minor histocompatibility antigens (recipient and sibling donor) unknown / pending for BOTH patient and sibling donor. HLA-identical siblings can still differ at HA-1/HA-2. (unknown); CD33 positive (blasts CD33+ by flow cytometry); CD123 positive (blasts CD123+ by flow cytometry); AML immunophenotype (supporting) CD34+ (increased), CD13+, CD15+ (subset), CD117+ (variably increased), HLA-DR (variably decreased), CD38 decreased; negative CD4/CD5/CD7/CD14/CD16/CD19/CD56/CD64; FLT3 negative / no mutation (2019 diagnostic NGS); IDH1 negative / no mutation (2019 diagnostic NGS); IDH2 negative / no mutation (2019 diagnostic NGS); JAK3 variant of uncertain significance (VUS) — 2019 (unknown); SMC3 variant of uncertain significance (VUS) — 2019 (cohesin-complex gene) (unknown); Cytogenetics / karyotype Complex, adverse-risk. 2019: complex karyotype with del(5q), del(7q), del(20q12), der(17)t(13;17) [17p], monosomy 13, others. 2026 relapse: gain of 5' MECOM, del(5q), del(7q), KMT2A amplification (4 copies), del(8cen), del(16q22). Shared del(5q)/del(7q) + TP53 founder across timepoints indicates evolution of the ORIGINAL clone transformed to AML (not a new/secondary or donor-derived leukemia). (ngs_pending)
- **Efficacy/toxicity weight:** 0.75

11 ranked therapeutic options across 6 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Allograft Hct2 interventions

1. Second allogeneic HCT (HCT2) for relapse after first allo-HCT

Likelihood of effect: Moderate in this subset - ~37% 2-year OS in matched-related-donor HCT2 (Christopeit 2013), lifted by her ~6-year first remission but pulled down by 85% blasts.

Toxicity burden: High (non-relapse mortality and acute/chronic GVHD substantial in a second allograft; infection; conditioning organ toxicity)

Counter-productive MoA:Moderate (Disease already escaped graft-versus-leukemia once; TP53-aberrant complex-karyotype biology predicts renewed immune escape after HCT2)

Overall:The curative backbone with a real cure fraction, but only deliverable after cytoreduction and a fitness workup that has not been done; disease burden at transplant dominates the odds.

Key references: PMID 23918951 · PMID 16110027 · PMID 17909197

2. Iomab-B (131I-apamistamab, anti-CD45 radioimmunotherapy conditioning) into HCT2

Likelihood of effect: Moderate - durable CR 17.1% vs 0% (SIERRA phase 3, EFS HR 0.23); ITT OS flat (HR 0.99) with 52% crossover; prior-HCT eligibility unverified.

Toxicity burden: High (grade >=3 treatment-related AEs ~60%: cytopenias, febrile neutropenia, infection, mucositis)

Counter-productive MoA:Low (Marrow-directed CD45 conditioning may spare the extramedullary thigh sarcoma sanctuary site, leaving a source of relapse; needs directed RT)

Overall:The RCT-grade conditioning enabler that could carry an 85%-blast marrow into HCT2, but not approved - the FDA declined the SIERRA package and prior-transplant eligibility for the go-forward study is unverified.

Key references: PMID 39298738 · NCT02665065 · NCT07157514

3. Donor lymphocyte infusion (DLI) from the sibling donor

Likelihood of effect: Low for this stratum - 21% vs 9% 2-year OS overall (Schmid 2007), but ~15% in the active-disease group she falls in at 85% blasts.

Toxicity burden: Moderate (acute/chronic GVHD is the characteristic toxicity; marrow aplasia when given into active disease)

Counter-productive MoA:Low (DLI into 85% blasts drives GVHD without meaningful graft-versus-leukemia; demands cytoreduction to remission first)

Overall:A readily available sibling-donor adjunct whose honest expectation is modest in florid disease; belongs after cytoreduction, alongside the transplant, not as standalone salvage.

Key references: PMID 17909197

4. FLAMSA-RIC sequential chemotherapy + reduced-intensity allo-HCT + prophylactic DLI

Likelihood of effect: Moderate - 88% CR and 42% 2-year OS in refractory / adverse-karyotype AML (Schmid 2005), matched-sibling subset included; delivers HCT2 in refractory disease.

Toxicity burden: High (GVHD and prophylactic-DLI-associated GVHD dominant; non-relapse mortality; conditioning organ toxicity)

Counter-productive MoA:Low (Prophylactic-DLI GVHD without proportional graft-versus-leukemia if disease burden is not reduced first)

Overall:The chemo-based conditioning alternative to Iomab-B for the same transplant job; a transplant-center choice, not an independent modality, with the prophylactic-DLI schedule dovetailing into consolidation.

Key references: PMID 16110027 · PMID 17909197

Evidence in detail

Second allogeneic HCT (HCT2) for relapse after first allo-HCT

Christopeit et al. *Journal of Clinical Oncology* 2013

- **Disease context:Hematologic relapse of acute leukemia after first allogeneic HCT** (2L+) — EBMT registry of 179 second transplants for relapse after a first transplant from matched related (n=75) or unrelated (n=104) donors; analysis of donor change. The patient's first HCT was from a matched sibling, so the related-donor subgroup is the relevant reference.
- **n:** 179
- **Toxicities:** Registry outcome study; per-term toxicity and non-relapse mortality are not broken out in the abstract. Remission duration after HSCT1 and disease stage at HSCT2 were the dominant survival drivers.
- **Efficacy:OS_rate** 25%; **OS_rate** 37%; **CR_rate** 74%

Schmid et al. *Journal of Clinical Oncology* 2005

- **Disease context:High-risk AML and MDS (mostly progressive/refractory disease)** (2L+) — 75 consecutive high-risk patients (59 progressive/refractory), 49% unfavourable karyotype; 31 with HLA-identical sibling donor. Fludarabine/cytarabine/amsacrine cytoreduction, then 4 Gy TBI/ATG/cyclophosphamide RIC, then prophylactic DLI from day +120.
- **n:** 75
- **Toxicities:** GVHD and prophylactic-DLI-associated GVHD are the principal risks; survival was best with high CD34+ cell dose and limited GVHD. Per-term toxicity is not tabulated in the abstract.
- **Efficacy:CR_rate** 88%; **OS_rate** 42%; **DFS** 40%

Schmid et al. *Journal of Clinical Oncology* 2007

- **Disease context:AML in first haematological relapse after allogeneic HCT** (2L+) — EBMT registry of 399 patients with AML in first relapse post-HCT; 171 treated with DLI vs 228 without. Multivariate analysis of survival drivers among DLI recipients.
- **n:** 399
- **Toxicities:** GVHD is the characteristic DLI toxicity; the abstract reports survival rather than per-term AE rates. Clinical benefit was limited to a minority — best when DLI was given in remission or with favourable karyotype, worst in aplasia or active disease.
- **Efficacy:OS_rate** 21%; **OS_rate** 9%; **OS_rate** 56%; **OS_rate** 15%

Iomab-B (131I-apamistamab, anti-CD45 radioimmunotherapy conditioning) into HCT2

Gyurkocza et al. *Journal of Clinical Oncology* 2025

- **Disease context:Active relapsed/refractory AML, age >=55, planned allogeneic HCT** (2L+) — Patients >=55 with active R/R AML randomised to a 131I-apamistamab-led regimen before alloHCT vs conventional care (with crossover to apamistamab allowed on non-response). Enrolled patients heading to a first alloHCT.
- **n:** 153
- **Toxicities:any treatment-related adverse event** (grade >=3): 59.7% (n=76); **any treatment-related adverse event** (grade >=3): 59.2% (n=77)
- **Efficacy:CR_rate** 17.1% (95% CI 9.4-27.5); **CR_rate** 0% (95% CI 0-4.7); **HR_OS** 0.99 (95% CI 0.7-1.41); **HR_EFS** 0.23 (95% CI 0.15-0.34)

Donor lymphocyte infusion (DLI) from the sibling donor

Schmid et al. *Journal of Clinical Oncology* 2007

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FLAMSA-RIC sequential chemotherapy + reduced-intensity allo-HCT + prophylactic DLI

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- **Efficacy:OS_rate** 21%; **OS_rate** 9%; **OS_rate** 56%; **OS_rate** 15%

Cd33 interventions

1. Gemtuzumab ozogamicin (Mylotarg, CD33 ADC) as cytoreductive bridge

Likelihood of effect: Low-to-moderate - ~30% CR+CRp in less-pretreated first relapse (Sievers 2001); likely lower post-FLAG-IDA/venetoclax; density-dependent, quantitative CD33 pending.

Toxicity burden: Moderate (grade >=3 hyperbilirubinemia 23%, transaminitis 17%, infection 28%; veno-occlusive disease class risk peri-transplant)

Counter-productive MoA:Low (Calicheamicin hepatotoxicity / VOD could compromise the very HCT2 it is meant to bridge to; time the interval accordingly)

Overall:The only agent a treating team can simply write; a modest on-label CD33 debulk whose VOD signal has to be timed around the planned second transplant.

Key references: PMID 11432892

2. Dual CD33/CD123 compound CAR-T (CD123-CD33 cCAR; CLL-1/CD33/CD123 CAR-T)

Likelihood of effect: Speculative - dual-antigen CAR-T outperformed single-antigen across four AML xenografts (Petrov 2018); zero human efficacy data, no post-allo-HCT experience.

Toxicity burden: Unknown - no clinical data (expected CRS and on-target myeloablation of normal CD33/CD123 progenitors)

Counter-productive MoA:Moderate (On-target myeloablation of normal CD33/CD123 progenitors compounds marrow aplasia in a re-transplant candidate; CRS; no post-HCT data)

Overall:The dual-antigen escape hedge for a clonally evolving relapse, held back by xenograft-only evidence and China-only trial access; gemtuzumab is the available CD33 fallback.

Key references: NCT04156256 · PMID 29515236

Evidence in detail

Gemtuzumab ozogamicin (Mylotarg, CD33 ADC) as cytoreductive bridge

Sievers et al. *Journal of Clinical Oncology* 2001

- **Disease context:CD33-positive AML in untreated first relapse** (2L+) — CD33+ AML in first relapse, no antecedent haematologic disorder, median age 61; three pooled open-label single-arm trials. Population is less heavily pretreated than this patient (no prior FLAG-IDA/venetoclax failure or extramedullary disease).
- **n:** 142
- **Toxicities:hyperbilirubinemia** (grade >=3): 23% (n=142); **hepatic transaminase elevation** (grade >=3): 17% (n=142); **infection** (grade >=3): 28% (n=142); **mucositis** (grade >=3): 4% (n=142); **severe nausea/vomiting** (grade >=3): 11% (n=142)
- **Efficacy:CR_rate** 30%

Cd123 interventions

1. Pivekimab sunirine (Decnupaz, IMGN632, CD123 ADC) as cytoreductive bridge

Likelihood of effect: Low - single-agent ORR 21% (95% CI 8-40), composite CR 17% at RP2D (Daver 2024); density-directed selection still pending.

Toxicity burden: High (reversible veno-occlusive disease, febrile neutropenia 10%; one treatment-related death; peri-transplant hepatic risk)

Counter-productive MoA:Moderate (Reversible VOD could compromise the very HCT2 it is meant to bridge to; density-dependent activity unconfirmed on qualitative flow)

Overall:A CD123 ADC that debulks without a failed backbone, but its boxed VOD warning works against the transplant it bridges to and on-label gemtuzumab covers the same job.

Key references: PMID 38423051 · NCT06561152 · PMID 29661755

Evidence in detail

Pivekimab sunirine (Decnupaz, IMGN632, CD123 ADC) as cytoreductive bridge

Daver et al. *Lancet Oncology* 2024

- **Disease context:Relapsed/refractory acute myeloid leukaemia, CD123-positive** (2L+) — CD123+ R/R AML, ECOG 0-1, one or more prior AML treatments; recommended phase 2 dose expansion n=29.
- **n:** 91
- **Toxicities:febrile neutropenia** (grade >=3): 10% (n=29); **infusion-related reaction** (grade >=3): 7% (n=29); **anaemia** (grade >=3): 7% (n=29); **veno-occlusive disease** (grade any): (n=68); **treatment-related death** (grade 5): 1% (n=68)
- **Efficacy:ORR** 21% (95% CI 8-40); **CR_rate** 17% (95% CI 6-36)

Hla A0201 Ha interventions

1. HA-1 / HA-2-directed donor TCR-T after HCT2 (Bleakley platform; TSC-100 / TSC-101, ALLOHA; BSB-1001) *(conditional on hla_a0201_ha positive)*

Likelihood of effect: Signal-level, positive branch - 4/9 CR with one remission >2 years and no DLTs (Bleakley n=9); feasibility, not efficacy-powered. Foreclosed if typing is negative.

Toxicity burden: Low (no DLTs across n=9; lymphodepletion cytopenias and GVHD are the setting risks; per-grade AE table not retrievable from the full text)

Counter-productive MoA:Low (Potential HA-antigen-loss escape; the graft-versus-leukemia-without-GVHD benefit depends on the mismatched mHAg staying hematopoietic-restricted)

Overall:The donor TCR-T aimed at the antigen axis the run was built around - highest ceiling, thinnest evidence, and hostage to a HA-1/HA-2 mismatch that genotyping the sibling pair will settle.

Key references: PMID 38683966 · NCT05473910 · NCT06704152

Evidence in detail

HA-1 / HA-2-directed donor TCR-T after HCT2 (Bleakley platform; TSC-100 / TSC-101, ALLOHA; BSB-1001)

Krakow et al. *Blood* 2024

- **Disease context:Persistent/recurrent AML, ALL, MDS or JMML after allogeneic HCT** (2L+) — HLA-A*02:01-positive recipients with recipient HA-1(H)-positive / donor HA-1-negative mismatch; disease recurrence after allo-HCT; donor-derived product after lymphodepletion.
- **n:** 9
- **Toxicities:dose-limiting toxicity** (grade any): (n=9)
- **Efficacy:CR_rate** 4/9 patients; **AE_rate** 0 DLTs

Hla A0201 interventions

1. WT1 HLA-A*02:01 TCR-T (FH-WT1-E50 / Greenberg WT1-TCRc4); A*24:02 WT1-siTCR fallback *(conditional on hla_a0201 positive)*

Likelihood of effect: Setting-mismatched, positive branch - 100% RFS at 44 months as post-HCT prophylaxis (Chapuis 2019, n=12, confounded comparator); unproven for florid relapse.

Toxicity burden: Low (engineered into EBV-specific cells to minimize GVHD; no per-term AE breakdown; on-target hematopoietic risk theoretical)

Counter-productive MoA:Moderate (Validated as relapse prophylaxis, not salvage; unproven against florid marrow disease; WT1 is a shared self-antigen)

Overall:The second HLA-restricted lever that keeps the immunotherapy axis open if HA-1/HA-2 does not mismatch, but its evidence is prophylaxis in MRD, not salvage of an 85%-blast marrow.

Key references: PMID 31235963 · NCT07645469 · PMID 28860210

2. CBX-250 (TCR-mimetic CG1/HLA-A*02:01 bispecific T-cell engager; CROSSCHECK-001) *(conditional on hla_a0201 positive)*

Likelihood of effect: Preclinical only, positive branch - sub-nanomolar killing and PDX survival benefit (ASH 2024 abstract); zero human efficacy readout. Foreclosed if A*02:01 is negative.

Toxicity burden: Unknown - first-in-human, no characterized AE profile (expected CD3-engager cytokine release syndrome, scaling with the 85%-blast burden)

Counter-productive MoA:High (Uncharacterized CRS scaling with 85%-blast burden could harm before it debulks; the standing conditional veto rests on this mechanism)

Overall:The only off-the-shelf option on the exact A*02:01 axis - fastest to move once typing confirms, but first-in-human with no efficacy and gated behind a cytoreduction-first CRS-mitigation plan.

Key references: NCT06994676 · doi:10.1182/blood-2024-201699

Evidence in detail

WT1 HLA-A*02:01 TCR-T (FH-WT1-E50 / Greenberg WT1-TCRc4); A*24:02 WT1-siTCR fallback

Chapuis et al. *Nature Medicine* 2019

- **Disease context:High-risk AML, relapse prophylaxis after allogeneic HCT** (maintenance) — HLA-A*02:01-positive, WT1-expressing high-risk AML; TCR inserted into EBV-specific donor CD8+ T cells and infused prophylactically post-HCT (not for active disease).
- **n:** 12
- **Toxicities:** Prophylactic product engineered into EBV-specific CD8+ T cells to minimise GVHD; abstract reports long-term persistence and polyfunctionality without a per-term AE breakdown.
- **Efficacy:RFS** 100%; **RFS** 54%

Tawara et al. *Blood* 2017

- **Disease context:Refractory AML and high-risk MDS** (2L+) — HLA-A*24:02-restricted WT1-specific TCR (retroviral vector with siRNA silencing of endogenous TCR); two dose-escalation cohorts, transfers 4 weeks apart followed by WT1 peptide vaccine.
- **n:** 8
- **Toxicities:on-target off-tumor normal-tissue toxicity** (grade any): (n=8)
- **Efficacy:other** 2/8 patients; **OS_rate** 4/5 patients

Tp53 Y220C interventions

1. Rezatapopt (PC14586, Y220C-selective p53 reactivator) + azacitidine *(conditional on tp53_y220c positive)*

Likelihood of effect: Cross-tumor only, positive branch - solid-tumor ORR 20% (30% KRAS-WT, PYNNACLE); no myeloid clinical data; preclinically needs a venetoclax partner.

Toxicity burden: Low (grade >=3 anaemia 16%; predominant low-grade nausea 58% and vomiting 44%, less frequent with food)

Counter-productive MoA:Moderate (Reactivated p53 barely triggers apoptosis alone - it needs a venetoclax / MDM2 partner the patient is already refractory to)

Overall:A mechanistically elegant Y220C longshot off the stated axis, double-gated on confirming TP53 then Y220C, with no myeloid data and a venetoclax dependency she has already failed.

Key references: PMID 40608889 · PMID 41740031 · NCT06616636

Evidence in detail

Rezatapopt (PC14586, Y220C-selective p53 reactivator) + azacitidine

Dumbrava et al. *New England Journal of Medicine* 2026

- **Disease context:Advanced/metastatic solid tumors harbouring TP53 Y220C** (2L+) — Heavily pretreated locally advanced/metastatic solid tumors with a TP53 Y220C mutation; PYNNACLE phase 1 dose escalation/optimization. Solid tumors only — no myeloid patients; responses restricted to KRAS wild-type tumors.
- **n:** 77
- **Toxicities:nausea** (grade any): 58% (n=77); **vomiting** (grade any): 44% (n=77); **blood creatinine increased** (grade any): 39% (n=77); **fatigue** (grade any): 39% (n=77); **anaemia** (grade any): 36% (n=77); **anaemia** (grade >=3): 16% (n=77); **treatment-related AE discontinuation** (grade any): 3% (n=77)
- **Efficacy:ORR** 20%; **ORR** 30%